System Acceptance Review

BRACU Mongol Tori
(revolution)

BRAC University

University Rover Challenge – 2018

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1. Introduction: BRACU Mongol Tori Revolution is a multidisciplinary team formed by a group of enthusiastic students of BRAC University. The rover is a fully functional, stand-alone mobile platform which is capable of acting as an astronaut assistant in various servicing and scientific research task. This year marks the third major update in the Mongol Tori lineup with numerous minor changes along the way. The major updates in the current version of the rover includes rover expansion system, switching from six wheeler to four wheeler design, reconstruction of the entire arm, portable command station and significant reduction in size and weight. The following report is about rovers' efficiency in terms of competing URC 2018 focusing on Core Rover Systems, Approach to Competition Tasks, Testing and Operations along with the unique features. The Mongol Tori project was divided into two distinct phase. The first being the development phase which lasted from August'17 till January'18. The testing phase began soon after, in February, and is expected to last till May'18.

2. Core Rover System: The entire rover system we came up with this year is inspired by our observation and learning form URC'17. Our main focus was to solve the difficulties faced in the previous year due to the overall construct of the rover. Therefore, we came up with quite a few changes this year.

2.1. Mechanical:

2.1.1. Chassis and Suspension: The chassis [fig 1(a)] of the rover has been designed using the combination of ladder and space frame structure. The K shape gives the chassis more rigidity and stiffness. The length of the whole chassis is 0.7 m and the width is 0.55 m. By installing triangulation into the chassis we have achieved the ability to diminish shape distortion due to unexpected shock from rocky terrain. The heavy space frame transmits the flexing load as tension and compression loads along the length of each strut. After a lot of experiment along with various suspension system, a four wheeler modified version of rocker bogie suspension has been incorporated in our rover. In this system there are two bogies directly connected with the chassis which is supported through a U shaped differential bar at the rear part of the rover. For maximizing the rotational angle of the bogies we have used universal joints in the differential bar. The unique feature of this system is the ability of expansion of the bogies [fig 1(b)]. The expanded suspension provides more stability during the vertical drops and slopes. The length of the bogies turns to 1.2 m from 0.76 m using two heavy duty linear actuators.

a b c d

Figure-1

2.1.2. Wheel: We improved the design of the wheel and came up with an optimal size and design by testing various types of custom-made wheels. The final version of the wheel [fig 1(d)] made from molded aluminum is very lightweight and strong which are 0.30m of diameter and a width of 0.10 meter. The distance between two wheels has been calculated in such a way so that the rover can have enough space for overcome obstacles. An efficient rubber grip has been used on the wheel for better traction. The orientation of the grip is fixed so that the rover can easily rotate 360 degree, climb rock and deal with the vertical drops.

2.1.3. Robotic Arm: A very strong and efficient robotic arm [fig 1(c)] with six degrees of freedom (base, shoulder, elbow, wrist, two degrees of freedom in the end effector) has been incorporated in our rover. The arm consists of four linear actuators and two dc motors. A bevel gear assembly has been assimilated at the tip of the arm, which gives the end effector more degrees of rotational freedom without wire twisting. A cable driven actuator system in the claw gives superior gripping flexibility. From the past years' experience we have built multiple end effectors focusing on specific task. For grabbing small elements we have designed a 3 finger claw with sufficient amount of rubber grip. An excavator claw based heavy duty end effector is developed for extreme retrieval and delivery task. The end effector can hold and the arm can lift more than 5 kg of weight at any circumstances.

2.1.4. Sample Collector: Two different sample collectors have been used in the rover. Both employing different means of collecting samples. The first sample collector uses a claw mechanism to dig and collect samples while avoiding outside contamination for as much as possible. This module is mounted at the end of the arm in place of the claw when needed. The other module uses a custom made drill bit housed inside a compact housing to ensure the soil be forced upward due to the drill bits rotation. The soil is then automatically collected from the top of the drill housing via a hole into a separate box. This entire drill based sample collection system is mounted on a separate actuator system.

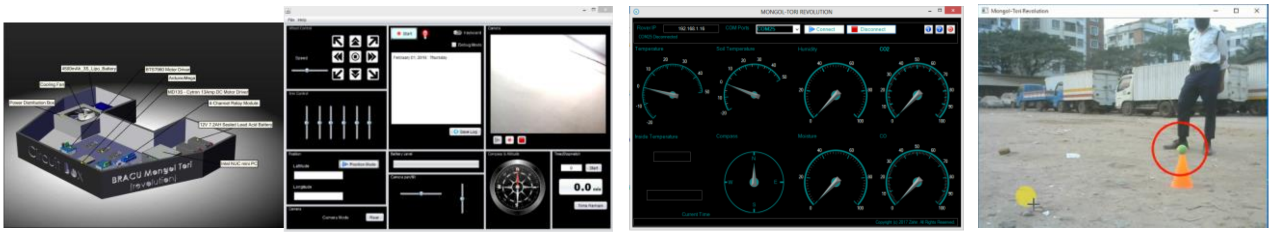
2.2. Control system:

This year we introduced a portable base-station that accommodates 3 laptops running control software and rover cameras to create a remote command station.

2.2.1. Electronics: ATmega2560 based Arduino microcontroller enables smooth control over the motors via 43A BTS7960 module. The microcontroller is directly connected with the on board NUC PC. We designed the entire electrical system using Proteus 8.0 and manufactured them on printed circuit board (PCB) which provides a convenient plug and play connection system. The above mentioned PCB consists of a total of six Cytron, four BTS motor driver, one four channel relay module and one Arduino mega [fig-2 (a)].

The main power source has been divided into two streams where one powers the motors and the other powers the communication system. 12V 30A Lead acid battery is used for the motors while two 12V 4.5A Li-Po batteries are used for the communication system. The branch powering the communication system further branches into four sub branches converting voltage using voltage converter which power the Switch, Router, IP camera, FPV camera and the on board NUC pc.

2.2.2. Software: We have developed several graphical user interfaces for control, science and mapping. The mapping GUI helps to track the rover in real time on a pre-loaded map using the latitude and longitude gotten form the GPS sensor. The software system establishes a very user friendly environment to control the rover. The systems' main aspect is to provide a reliable and easy to use control interface of the rover and to retrieve accurate up to date information of the rovers' status. We have developed the control GUI [fig-2 (b)] using JAVA programming language. In this system the base station acts as the client while the rover acts as the server and the protocol followed in the communication is Transmission Control Protocol (TCP). The science GUI [fig-2 (c)] developed in C# interprets the sensor readings into custom gauges which offer a graphical perception that makes it easy to distinguish between different states of perception. The autonomous part of the software is written using Python3.6 programming language and OpenCV3.0 image processing libraries. The software combines both GPS and magnetometer values in coordination with the visual data from the camera to achieve full autonomy [fig-2 (d)].



a b c d

Figure- 2

2.3. Communication: In order to establish a strong communication between base station and rover we used two routers of 2.4 GHz band with the ability to switch among frequency channels. Also routers with dual band (2.4GHz and 5.8GHz) support is planned to be implemented in the near future. Onboard NUC PC communicates with the base station over 2.4 GHz routers which are Rocket M2 and Bullet M2. For antenna, an Omni-directional antenna (TL-ANT2412D) has been used on rover to establish a 360 degree horizontal beam pattern to ensure more coverage. On the base station side a High gain directional sector antenna (ANT2327D15T-120DP) has been used with a Rocket M2. A rotator (G-5500) is used with this directional sector antenna to manually track the rover. The rotator is controlled from the base station with the elevation azimuth controller. For vision we have used two IP cameras and two FPV cameras with 5.8 GHz 600mW transmitter. Switching between FPV cameras can be done from the control software.

3. Approach to Competition Tasks: We arranged a mock competition to practice completing all the tasks in time. From the experience of URC'17 several similar representing locations were selected for the practice sessions. All the tasks are scheduled to be performed in a certain time limit.

3.1. Extreme Retrieval and Delivery Task:

Our rover is designed to travel up to 1 km from base station and convey help to astronauts. We have implemented a modified version of rocker bogie suspension system. The additional rubber grip on the aluminum wheels help to increase friction. The customized wheel motor assembly helps the motor to isolate from potential obstacle. This gives us the advantage to traverse through rocks, vertical drops of more than 0.75 m, slopes and stairs. The speed of the rover is approximately 4 kmh.

3.2. Equipment Servicing Task:

Our rovers' arm can reach a height of 1.5m from the ground having six degrees of freedom. The arm is utilized for Equipment Servicing, Science Cache, Extreme Retrieval and Delivery errands. Moreover, the arm is implemented with a very strong three finger claw. A rotational apparatus construct framework has been included with respect to the claw for 360 degree pivot without wire curving. For vision, a camera is fixed at the claw region of the arm. A box is attached with the rover to keep the tools and hardware after retrieval. The arm is able to carry up to 5kg of weight.

3.3. Autonomous Traversal Task:

Coming to the implementation of the autonomous traversal, we are using a U-blox 6 based NEO-6 series GPS module, an HMC5883L three-axis magnetic module and computer vision to achieve complete autonomous traversal. Using a GPS module we calculate the latitude and longitude with a 5 meter error margin which is experimented in open environment with preloaded map. The Compass sensor is used along with the GPS module to find out the exact rover direction. We use several target gates to get the expected traversal path from where the distance and the angle between start and every end positions were calculated accordingly. The image processing sub-routine takes over once the rover comes within 3.5 meter radius of the destination dictated by the GPS values. Within this radius the autonomy of the rover is maintained by the image processing routine constructed using OpenCV library and python programming language. After getting the values from the GPS sensor the software calculates the distance to destination and travels accordingly. Evaluating the data from the magnetometer it aligns the rover in the direction of the determined travel path. Once within range of the destination the software switches to image processing procedure for autonomous traversal.

4. Safety measures: An affair which is strictly being followed here is safety. First of all, our Rover has a "kill switch" which will immediately force shutdown the rover mechanisms when needed. Furthermore, to ensure the safety of our main circuit board we also used a 30 Amp fuse box.

5. Testing & Operation

Several experiments were conducted prior to the competition to ensure proper functionality of the rover. The experiments can be split into three steps:

a. Sub-System Testing:

The subsystems are wheels, suspension & chassis, end effector, sample collector and software. On 08/10/2017 we had made a custom ramp for testing different types of wheels. After selecting the wheel we had our indoor test on our rover chassis and suspension test on 22/10/2017. Considering different types of gear mechanism we have selected bevel gear assembly for our end effector and tested the module on 05/11/2017 by grabbing small stuffs like screwdriver, hammer etc. in our lab. Since we got a good result in URC-17 by using the drill mechanism for sample collection, this year we are considering a new excavator claw mechanism [fig 3(d)] alongside a modified version of the drill mechanism [fig 3(a)]. To bring some improvement we will construct the drill mechanism after SAR and by this time we have conducted a test of the excavator claw on 19/11/2017 inside our laboratory on an artificial soil pot. Moreover, we have developed a platform independent control GUI which is compatible with any system and rover. By getting this advantage we have tested our control GUI on a prototype rover on 02/01/2018.

b. Complete System Testing:

The communication and electronics system used in URC-2017 performed quite well so we have decided to build upon the same systems. For getting a positive confirmation with the new updated communication and electronics system the full setup was tested on the rover. It was done at the very beginning of our work for URC-2018. Moreover, we had participated Indian Rover Challenge (IRC)-2018 on 06/01/2018 with these systems and secured the second runners up position. So in one way this was a good practice before URC-2018. After preparing the total robotic arm system and different types of end effectors we have tested the full arm system on 04/11/2017 by picking up different objects.

c. Operational Testing

For operational testing purpose of our complete rover system we focused on the main four tasks of URC-2018. We have chosen a remote place where the surface was quite similar to the main surface of competition ground. In this place we had a point for vertical drop, steep slope and different types of rough terrain. We went to this test site on 10/01/2018 to test our rover with its complete system to perform the extreme retrieval and delivery task. For testing the equipment servicing task we have made a dummy setup and tested our rover inside our laboratory on 12/01/2018. In this test we have practiced power cord plug-in, plug-off from a panel, turning knob and undoing a latch etc. For the autonomous test, we chose another location where the GPS data were available. For a complete test of autonomous task on 11/02/2018 we have selected some points on the basis of GPS values and successfully completed the task. While we were doing our extreme retrieval and delivery task we collected some soil with our rovers' sample collector to perform the science task and lab tests in our science laboratory. Inside the laboratory on 20/02/2018 we have tested the soil in hopes of getting traces of life and we have successfully done microbial biomass test, water flow capillary test and amino acid test.

8. Science Plan:

The Science task team worked following a specific schedule to accomplish the challenges and evaluate the parameters which can define the environmental condition and detect sign of life as Mars is the most habitable planet after Earth. With the new rover BRACU Mongol-tori, we are focusing on searching the likelihood to support life on the red planet. The team has decided to perform onboard and lab experiments to verify the atmosphere of Mars. The rover will take a high-resolution panorama image of the site with GPS location data. The sample collector will drill [fig 3(a)] and collect the samples from the soil in an airtight container for further lab tests. Mongol Tori is designed to perform onboard tests for subsurface data collection using CO Sensor (MQ7), CO2 Sensor (MQ135), H2 (MQ8) & O2 (GROVE gas sensor) Sensor, Light Intensity sensor (LDR), UV sensor (ML8511), Atmosphere Temperature Sensor (DHT22), Air Humidity Sensor (DHT22), Soil Temperature Sensor (DS18B20), Soil Moisture Sensor. Along with that it will take a high resolution close up image of the soil for stratigraphic profile. After collecting the samples the rover will return to the base station. At the Lab, Microbial Biomass Test, Water Flow Capillary Test, pH Test [fig 3(b)], Amino Acid Test, Microscopy of Soil Particles [fig 3(c)] and Soil Bulk Density Test will be performed. The pH test indicates the Acidity or Basicity of the soil sample. Then from microbial biomass test we can determine the amount of biomass (Carbon, Nitrogen) present in soil using the following equation.

Biomass = (difference of weight/previous weight) * 100

This will give us the possibility of components which form life. After that, we will perform water flow capillary test which shows us the possibility of water flow. Finally, we will put the samples under microscope and analyze the composition of soil for presence of water and mineral crystals. For the whole Science task the used lab equipment are - Test Tubes, Centrifugal Tube, Weight Machine, Digital Microscope, Spirit Burner, Burner Stand, Beakers and Glass-slides in our Natural Science Lab with the Reagents- Spirit/Ethanol, Water and Ninhydrin reagent. All the experiments individually represent the condition for living in Mars.

From the previous experiences and results of URC-2017 this year we have designed a new sample collector for URC-2018. By using this sample collector our rover can go to the desired spot and come back with sufficient amount of sample in a fixed time. The above mentioned lab tests were performed on the samples we have collected through this sample collector.

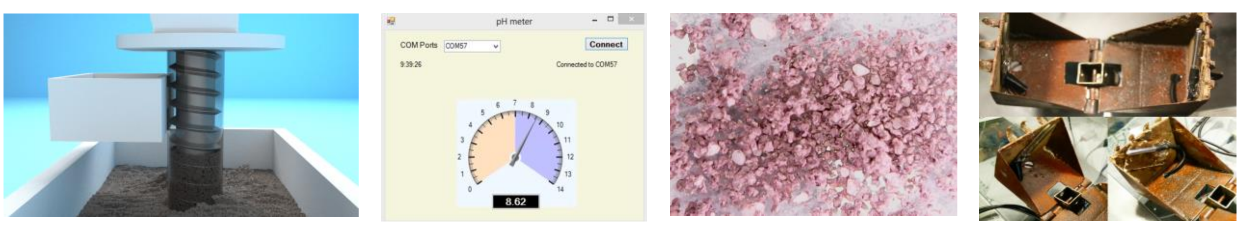


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Figure- 3

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